Quadratic variance occupation for Gaussian Fock formulas

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26 October 2026

Abstract

For equal-diagonal Gaussian tanh feature maps in compatible coordinates, we construct exact unbiased scalar arithmetic formulas with trace second moment at most two and expected coordinate size $CT_0 \exp[C \sum_j q_j^2]$. Higher Hermite degrees are created rarely, with probability independent of the physical variance q_j; their small coefficients supply the quadratic variance charge. Compatible Hilbert–Schmidt perturbations add their norm sum to the exponent. For a free near-critical Gaussian chain this gives the feature-representation scale $\exp[O(1 + \delta^2 H)]$. This is not a posterior learning theorem, and no variable-diagonal or non-Gaussian extension is asserted.

1 Feature map and precise scope

Let H be a real separable Hilbert space with a fixed countable coordinate basis. For $q > 0$, put $F(q) = \mathbb{E} \tanh^2(\sqrt{q}Z)$, $Z \sim N(0,1)$, and expand in orthonormal probabilists' Hermite polynomials:

$$\tanh(\sqrt{q}Z) = \sum_{k \text{ odd}} a_k(q) H_k(Z), \qquad b_k(q) = a_k(q)/\sqrt{F(q)}.$$

For $\|u\| = 1$, define the normalized Fock feature

$$G_q(u) = \bigoplus_{k \text{ odd}} b_k(q) u^{\otimes k}.$$

Full ordered tensor coordinates are used. Parseval gives $\|G_q(u)\| = 1$, and the Gaussian Hermite identity gives

$$\langle G_q(u), G_q(v) \rangle = \sum_k b_k(q)^2 \langle u, v \rangle^k = \frac{\mathbb{E} \tanh(\sqrt{q}Z_u) \tanh(\sqrt{q}Z_v)}{F(q)},$$

where the standard Gaussian pair has correlation $\langle u, v \rangle$. Thus the common q is a physical preactivation variance, equal at every query. It is not a query-dependent scalar oracle.

Theorem 1. *On a finite input set X, consider finitely many steps G_{q_j}, optionally interspersed with linear maps $J_j + K_j$. Each J_j is a signed partial coordinate injection (no duplicated old coordinates), and K_j is Hilbert–Schmidt in the same compatible bases. Assume all true intermediate states have norm at most one, and every state entering G_{q_j} has norm exactly one. Suppose initial coordinates have unbiased random scalar formulas with pointwise trace second moment at most two and common expected size $T_0 \geq 2$. Then exact unbiased coordinate formulas satisfy*

$$\sum_i \mathbb{E} Y_i(x)^2 \leq 2 \quad (x \in X), \qquad \sup_i \mathbb{E}\, \mathrm{size}(Y_i) \leq CT_0 \exp[C(S + Q_2)],$$

where $S = \sum_j \|K_j\|_{\mathrm{HS}}$ and $Q_2 = \sum_j q_j^2$. The random shapes are independent of the query, and the constants are universal. In particular $S = 0$ for an unmodified Fock chain.

Formulas here are finite real arithmetic trees with arbitrary real edge coefficients. A copied coordinate changes its final edge coefficient; successive coefficients are combined algebraically. A zero is one leaf. This prevents a new unary node at every skipped layer. Converting a final weighted tree into ordinary scalar multiplications costs only a constant per edge. No finite-bit or computational-complexity conclusion is needed for the theorem. An arbitrary dense basis rotation is not a free copy.

2 The exact Hermite estimate and scalar factory

Lemma 2. *For every $q > 0$, $\sum_{k\geq 3,\ k \text{ odd}} k^2 b_k(q)^2 \leq 6q^2$.*

Proof. Gaussian Sobolev Parseval and bounded derivatives give

$$\sum_k k(k-1)a_k(q)^2 = q^2\mathbb{E}[\tanh''(\sqrt{q}Z)^2].$$

For $k \geq 3$, $k^2 \leq \frac{3}{2}k(k-1)$. Also $\tanh'' z = -2\tanh z\,\operatorname{sech}^2 z$, so $\tanh''(z)^2 \leq 4\tanh^2 z$. Division by $F(q) > 0$ proves the claim. It is an exact all-degree estimate, not a Taylor truncation. □

Suppose current coordinate oracles X_i have means u_i, $\|u\| = 1$, pointwise trace second moment at most $V \geq 1$, and common expected size at most $T \geq 2$. At an odd degree $k \geq 3$, set

$$m_k = \lceil kV \rceil, \qquad p_k = (C_* k^2 V)^{-1}.$$

For an ordered word $I = (i_1, \ldots, i_k)$, return zero with probability $1 - p_k$, and otherwise return

$$\frac{b_k(q)}{p_k} \prod_{a=1}^{k} \overline{X}_{i_a},$$

where each bar averages m_k independent child copies. Distinct factors are independent even if their coordinate indices coincide. The oracle is unbiased. Unit norm of the true mean gives

$$\sum_i \mathbb{E}\overline{X}_i^2 \leq 1 + \frac{V-1}{m_k}, \qquad \left(\sum_i \mathbb{E}\overline{X}_i^2\right)^k \leq e.$$

There is a universal arithmetic-accounting constant C_0 for which the nonzero branch has expected size at most $C_0 k m_k T$. Since $m_k \leq 2kV$, the whole coordinate costs at most

$$1 + C_0 p_k k m_k T \leq 1 + (2C_0/C_*)T \leq T$$

when $C_* \geq 4C_0$. At degree one return the coefficient-weighted copy $b_1(q)X_i$, also of size at most T.

Tonelli's theorem and independence give the degree-k trace bound $e b_k(q)^2/p_k$. Summing and applying Lemma 2 yields

$$V_{\text{new}} \leq b_1(q)^2 V + e\sum_{k\geq 3} \frac{b_k(q)^2}{p_k} \leq V(1 + C_1 q^2), \qquad T_{\text{new}} \leq T. \tag{1}$$

All sums are nonnegative, so countably many coordinates cause no exchange of conditionally convergent series. The higher-degree coordinates are new orthogonal coordinates: their zero branch does not evaluate an old coordinate. This is why p_k can be independent of q. Averaging before forming the tensor product also avoids the potentially divergent naive series $\sum_k b_k(q)^2 V^k$.

3 Variance resets and compatible controls

Start by averaging two copies if needed, so the trace bound is at most $3/2$, at constant size cost. For a small step with $C_1 q^2 \le 1/8$, update the deterministic bound by (1). If it first exceeds $7/4$, it is still below two. Average two independent copies of each coordinate. Since the true mean has norm one, the trace becomes at most $(V_{\mathrm{new}} + 1)/2 \le 3/2$, at size multiplier at most three. Between these resets the trace must increase by at least $1/4$, while each increase is at most Cq^2. Thus there are at most $CQ_2 + C$ such resets. A large step starting from $V \le 2$ has $V_{\mathrm{new}} \le 2(1 + C_1 q^2)$. Averaging $\lceil 2V_{\mathrm{new}} \rceil$ copies restores trace at most $3/2$. Its size multiplier is polynomial in $1 + q^2$, with logarithm $O(q^2)$ above the fixed small-step cutoff. This proves the theorem when $S = 0$.

For completeness, the compatible linear-control estimate has the same reset form. Let $\kappa = \|K\|_{\mathrm{HS}}$. In row i put $c_i = \sum_j K_{ij}^2$ and sample j with probability K_{ij}^2/c_i. The oracle $Z_i = K_{ij} X_j / (K_{ij}^2/c_i)$ has mean $(Ku)_i$ and

$$\mathbb{E} Z_i^2 \le c_i V, \qquad \sum_i \mathbb{E} Z_i^2 \le \kappa^2 V.$$

Zero rows are omitted. For small nonzero κ, add this row oracle to the copied J coordinate only on a Bernoulli branch of probability $p = \kappa$, dividing its coefficient by p. Fresh row and copy children are independent. The mean is $(J + K)u$; its trace is bounded by

$$V + 2\|Ju\|\|Ku\| + \kappa^2 V/p \le V + C\kappa$$

while $V \le 2$ and $\|u\| \le 1$. Common expected size grows by at most $1 + C\kappa$, since the zero branch is a literal coordinate copy. For zero κ no branch is called. Above the fixed small-control cutoff use $p = 1$ and an averaging reset; trace and size costs are polynomial in $1 + \kappa$, with logarithm $O(\kappa)$. When an intermediate true mean has norm below one, averaging m copies still gives trace at most $1 + (V - 1)/m$. Resets can therefore be charged to the combined increments $C(\kappa + q^2)$. Multiplying the costs proves Theorem 1 with exponent $C(S + Q_2)$.

Every realized tree is finite almost surely: there are finitely many trajectory layers, every selected tensor degree is finite, and every averaging count is finite. No vector subcomputation is shared.

4 Readouts, near-critical scale, and limitations

For $v \in \ell_2$, sample coordinate i with probability $v_i^2/\|v\|^2$ and return $\|v\|^2 Y_i / v_i$. This is an unbiased oracle for $\langle v, h(x) \rangle$, with second moment at most $2\|v\|^2$ and the same common expected size bound. The zero readout is trivial. If the readout has global signed margin one on X, average $O(1 + \|v\|^2)$ independent copies per group. Chebyshev's inequality makes a group accurate to $1/2$ with probability at least $3/4$. Use $O(\log(2|X|))$ independent groups; concentration and a union bound make a majority correct at every input with positive probability. Markov's inequality simultaneously bounds total formula size by a constant times its expectation. Thus a fixed-seed classifier exists as a majority of signs of real arithmetic formulas of total size

$$\mathrm{poly}(\log(2|X|), T_0, 1 + \|v\|) \exp[C(S + Q_2)].$$

This exact real-constant statement is separate from any finite-bit completion.

For an uncontrolled equal-diagonal Gaussian tanh chain with $t = 1 + \delta$, $0 \le \delta \le \delta_0$ for a fixed small δ_0, and bounded initial variance, the free variance comparison gives $q_j \le C[\delta + (j + 1)^{-1}]$. Indeed $F(q) \le q - cq^2$ on a fixed bounded variance interval: $(q - F(q))/q^2$ is positive and continuous, with limit two at zero. Thus $q_{j+1} = t^2 F(q_j) \le (1 + C\delta) q_j - c q_j^2$. For a sufficiently large fixed K, the interval $[0, K\delta]$ is invariant when δ_0 is small. Above this interval the sequence obeys $q_{j+1} \le q_j - (c/2) q_j^2$,

so its reciprocal increases by at least $c/2$ until it enters the interval. This proves the comparison. Consequently

$$Q_2 = \sum_{j<H} q_j^2 \le C(1 + \delta^2 H).$$

The normalized Gaussian Fock feature/readout representation therefore has resource $\exp[O(1 + \delta^2 H)]$, multiplied by the stated initial and readout resources. This identifies a quadratic free-feature timescale; it is not a learning statement for a margin-conditioned prior.

The common q_j and compatible bases are substantive hypotheses. Query-dependent variances require input-dependent Hermite coefficients; the known variance-oracle and normalization helpers still have an $O(q_j)$ charge. An arbitrary dense change of basis is not a signed coordinate copy. Margin conditioning may change Gaussian row laws, and physical readout normalization may change its norm. Nothing here provides a variable-diagonal or non-Gaussian posterior bridge, or settles the original canonical Boolean tanh depth question.